

Question Bank for Basic Electronics (BBEE103/203)

Module – 2

Bipolar Junction Transistors

1. With a neat diagram, explain the operation of an *npn* transistor.
2. Explain various currents and voltages flowing through the BJT transistor. *(MQP '22 – 7M)*
3. Define α and β . Determine the relationship between α and β . *(MQP '22 – 6M)*
4. A transistor has $\beta = 150$ and I_E is 12 mA. Calculate the approximate collector current (I_C) and base current (I_B). *(Feb '23 – 4M)*
5. Calculate α_{dc} and β_{dc} for the transistor if I_C is measured as 1 mA and I_B is 25 μ A.
6. Describe with neat circuit diagram of BJT amplification for both voltage and current. *(Feb '23 – 8M)*
7. Explain BJT current amplification for increasing and decreasing I_B levels. *(MQP '22 – 8M)*
8. Describe how a transistor can be used as a voltage amplifier. *(MQP '22 – 4M)*
9. Explain input and output characteristics of the common emitter configuration. *(MQP '22 – 6M)*
10. Briefly explain common collector input characteristics. *(Feb '23 – 4M)*
11. Draw the DC load line for transistor and identify Q point. *(MQP '22 – 8M)*

Field Effect Transistors

1. Explain the construction and operation of N-channel JFET. *(Feb '23 – 8M)*
2. Explain the transfer and drain characteristics of a JFET.
3. Make use of N-channel enhancement type MOSFET and describe the construction and working. *(Feb '23 – 7M)*
4. Explain the enhancement type MOSFET along with the drain characteristics. *(MQP '22 – 8M)*
5. Explain the transfer and drain characteristics of enhancement type MOSFET.
6. Explain depletion type MOSFET along with the transfer and drain characteristics. *(Feb '23 – 8M)*

Module – 3

Operational Amplifiers

1. What is an operational amplifier? Mention its applications.
2. Explain block diagram representation of typical op-amp. *(Feb '23 – 8M, MQP '22 – 6M)*
3. Describe block diagram representation of an op-amp. Also describe its operational behavior with an equivalent circuit. *(MQP '22 – 8M)*

4. Explain the following op-amp parameters
(i) CMRR (ii) Slew rate (iii) Input offset voltage (iv) Input bias current (Feb '23 – 8M)
5. With respect to an op-amp, explain the following:
(i) Input offset voltage (ii) Slew rate (MQP '22 – 8M)
6. Define Op-Amp parameters: Gain, CMRR, Slew Rate, Input resistance. (MQP '22 – 8M)
7. Explain the following terms related to op-amp: (i) Open loop voltage gain (ii) Common mode gain (iii) CMRR (iv) Maximum Output Voltage Swing (v) Input Offset Voltage (vi) Input Offset Current (vii) Input bias current (viii) Input resistance (ix) Output resistance (x) Slew rate (xi) PSRR/Supply voltage rejection ratio (xii) Virtual ground.
8. A certain op-amp has an open loop voltage gain of 1,00,000 and a common mode gain of 0.2. Determine the CMRR and express it in decibels.
9. An op-amp has a slew rate of $0.8 \text{ V}/\mu\text{sec}$. What is the maximum amplitude of undistorted sine wave that the op-amp can produce at a frequency of 40 kHz? What is the maximum frequency of the sine wave that op-amp can reproduce if the amplitude is 3 V?
10. Briefly discuss the ideal characteristics of the op-amp. (MQP '22 – 7M)

Op-Amp Applications

1. Explain op-amp as an inverting and non-inverting amplifier. (Feb '23 – 8M)
2. Explain inverting amplifier. (MQP '22 – 4M)
3. An inverting amplifier using op-amp has a feedback resistor of $10 \text{ k}\Omega$ and one input resistor of $1 \text{ k}\Omega$. Calculate the gain of the op-amp and the output voltage if it is supplied with an input of 0.5 V . (MQP '22 – 4M)
4. An inverting amplifier has $R_1 = 20 \text{ k}\Omega$, $R_f = 100 \text{ k}\Omega$. Find the output voltage, input resistance and input current for an input voltage of 1 V .
5. A non-inverting amplifier has closed loop gain of 25. If input voltage $V_i = 10 \text{ mV}$, $R_f = 10 \text{ k}\Omega$, determine the value of R_1 and output voltage V_o .
6. Explain working of a differential amplifier. (MQP '22 – 8M)
7. Briefly explain op-amp as a voltage follower. (Feb '23 – 4M)
8. Derive an expression for the 3-input summing circuit. (MQP '22 – 6M)
9. Describe a summing amplifier using an op-amp in an inverting configuration with three inputs. (MQP '22 – 8M)
10. Calculate the output voltage of a three-input inverting summing amplifier, given $R_1 = 200 \text{ k}\Omega$, $R_2 = 250 \text{ k}\Omega$, $R_3 = 500 \text{ k}\Omega$, $R_f = 1 \text{ M}\Omega$, $V_1 = -2 \text{ V}$, $V_2 = -1 \text{ V}$ and $V_3 = +3 \text{ V}$.
11. Design an adder circuit using an op-amp to give the output $V_o = -(3V_1 + 4V_2 + 5V_3)$. Assume $R_f = 120 \text{ k}\Omega$. (Feb '23 – 4M)

12. Design the circuit diagram for the output voltage $V_o = -5(0.1V_1 + 0.2V_2 + 10V_3)$. Also draw

- the neat circuit diagram. **(MQP '22 – 7M)**
13. Develop a summer circuit using op-amp to get the following output voltage $V_o = -(2V_1 + 2V_2)$. **(MQP '22 – 4M)**
14. Design an adder circuit using op-amp to obtain an output voltage, $V_o = -[2V_1 + 3V_2 + 5V_3]$. Assume $R_f = 10\text{ k}\Omega$. With a neat circuit diagram, explain the subtractor using an op-amp.
15. Derive an expression for integrator and differentiator. **(MQP '22 – 6M)**
16. Describe an integrating amplifier using an op-amp in an inverting configuration. **(MQP '22 – 8M)**
17. Explain op-amp as an integrator circuit with a neat input and output waveforms using square wave as input. **(Feb '23 – 8M, MQP '22 – 6M)**